

# AT LEAST 10 MILLION PIECES OF ARTIFICIAL DEBRIS ARE NOW IN EARTH ORBIT

## KEY

The image shows 22,300 objects monitored by the ESA Space Debris team by radar and telescopes.

-  **Satellites—mostly dead. About 2,300 operational**
-  **Spent rockets**
-  **Mission waste (nuts, gloves, lost items)**
-  **Debris from explosions and collisions**

## Low Earth Orbit

This region is full of orbiting spacecraft, but also full of waste material ejected from spacecraft during missions and countless pieces of debris from collisions.

## GPS (Global Positioning System) satellite

One of 31 forming a network, the GPS satellites orbit in one of six orbits. Each orbit is at a different angle to ensure they cover the entire surface of Earth. Someone on the ground is in contact with at least six of them at any one time.

## How high are satellites?

Most objects launched into space are in Low Earth Orbit (LEO). At the lowest LEOs (99 miles / 160 km) objects circle Earth in 87 minutes at 17,470 mph (28,100 kph). Certain orbits are particularly useful. Image-taking satellites use polar sun-synchronous orbits, which pass the equator at the same local time on every pass, so the shadows are the same.

## Geosynchronous orbit

22,236 miles (35,786 km)  
Satellites at this height orbit at the same speed as Earth turns, so they stay in the same spot over Earth's surface.

## Hubble Space Telescope

345 miles (555 km)

## Polar sun-synchronous satellites

373–497 miles (600–800 km)

1,244 miles  
(2,000 km)

HIGH EARTH  
ORBIT ZONE

MEDIUM EARTH ORBIT ZONE

LOW  
EARTH  
ORBIT  
ZONE

## GPS satellites

12,600 miles (22,200 km)  
Objects orbit once every 12 hours, or twice a day.

## International Space Station

255 miles (410 km)

